Review for chapter 11

Geometry in Space and Vectors

- **11.1 Cartesian Coordinates in Three-Space**
  - Cartesian Coordinates （笛卡尔坐标）
  - Points in three space
  - Distance
  - Midpoint(中点) formula
  - Surface and Curve in three-space
- **11.2 Vectors**
  - Definition
  - Operations（运算法则）
  - Unite vector（单位向量）

11.1 Cartesian Coordinates in Three-Space

- The Distance Formula: The distance between $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ is

$$\boxed{|P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}}$$

- the standard equation of a sphere

If (x, y, z) is a point on the sphere of radius r centered at (h, k, l), then

$$\boxed{(x-h)^2 + (y-k)^2 + (z-l)^2 = r^2.}$$

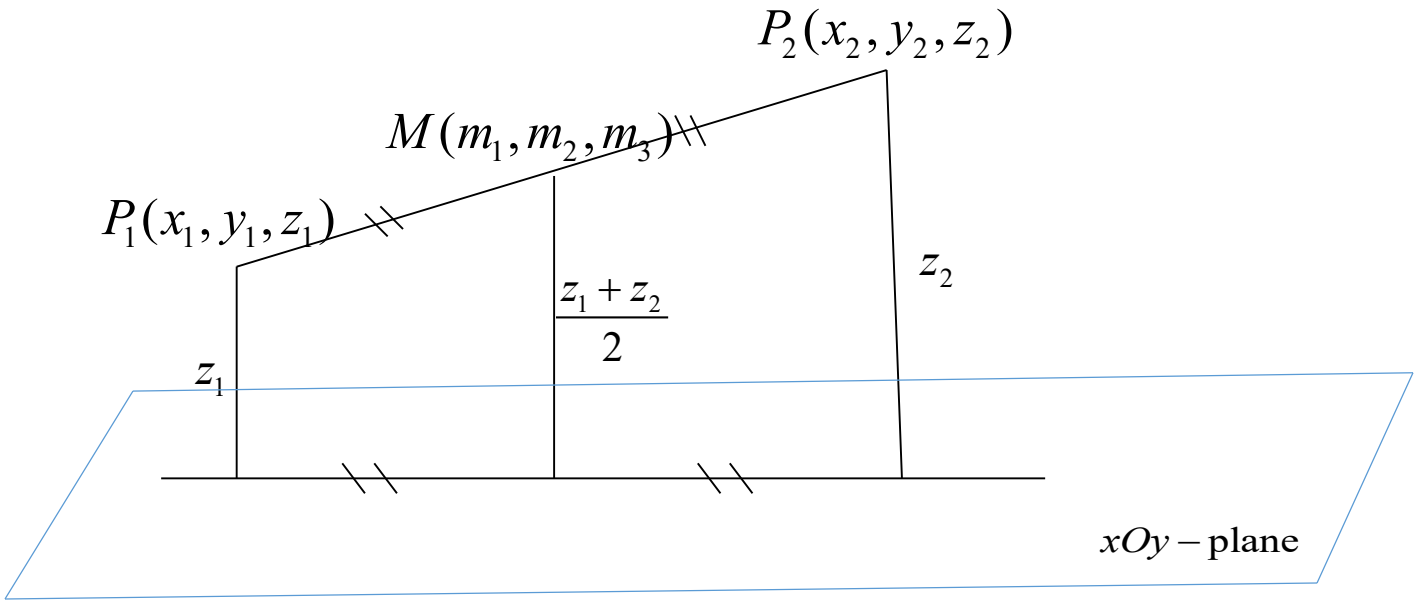
This is called *the standard equation of a sphere*.（球面的标准方程）

Midpoint (中点) Formula

If $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ are end points of a line segment, then

the midpoint $M(m_1, m_2, m_3)$ has coordinates

$$\boxed{m_1 = \frac{x_1 + x_2}{2}, \quad m_2 = \frac{y_1 + y_2}{2}, \quad m_3 = \frac{z_1 + z_2}{2}}$$



Curves in Three-Space A curve in three-space is determined by the parametric equations

$$x = f(t), \quad y = g(t), \quad z = h(t); \quad a \le t \le b$$

The arc length of this curve is

$$L = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2 + [h'(t)]^2}\, dt$$

11.2 Vector（向量）

Addition

$$\mathbf{u}+\mathbf{v}=\langle u_1+v_1, u_2+v_2, u_3+v_3\rangle$$

Scalar multiplication

$$c\mathbf{u}=\mathbf{u}c=\langle cu_1, cu_2, cu_3\rangle$$

Subtraction

$$\mathbf{u}-\mathbf{v}=\mathbf{u}+(-1)\mathbf{v}=\langle u_1-v_1, u_2-v_2, u_3-v_3\rangle$$

Three special vectors in three-space are

$$\mathbf{i}=\langle 1,0,0\rangle, \mathbf{j}=\langle 0,1,0\rangle, \mathbf{k}=\langle 0,0,1\rangle$$

These are called the standard unit vectors, or basis vectors.

Every vectors u can be written in terms of i,j,k as follows:

$$\mathbf{u}=\langle u_1, u_2, u_3\rangle=u_1\mathbf{i}+u_2\mathbf{j}+u_3\mathbf{k}$$

- Decomposition of a vector

Three special vectors in three-space are

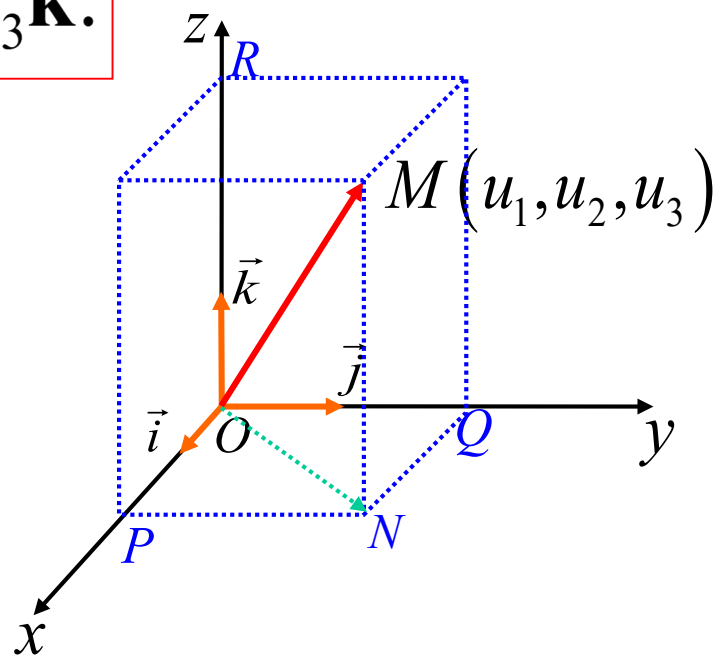
$$\mathbf{i} = \langle 1,0,0 \rangle, \mathbf{j} = \langle 0,1,0 \rangle, \mathbf{k} = \langle 0,0,1 \rangle$$

These are called the standard unit vectors, or basis vectors.
Every vectors u can be written in terms of i, j, k as follows:

$$\boxed{\mathbf{u} = \langle u_1, u_2, u_3 \rangle = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}.}$$

The magnitude of a vector is just the length of the arrow that represents it.

$$\|\mathbf{u}\| = \sqrt{u_1^2 + u_2^2 + u_3^2}$$



➢11.3 The Dot Product(点积)

$$\boxed{\mathbf{u}\cdot\mathbf{v}=\langle u_1,u_2,u_3\rangle\langle v_1,v_2,v_3\rangle=u_1v_1+u_2v_2+u_3v_3}$$

Theorem A Properties of the Dot Product

If u, v and w are vectors, and c is a scalar, then

1. $\mathbf{u}\cdot\mathbf{v}=\mathbf{v}\cdot\mathbf{u}$
2. $\mathbf{u}\cdot(\mathbf{v}+\mathbf{w})=\mathbf{u}\cdot\mathbf{v}+\mathbf{u}\cdot\mathbf{w}$
3. $c(\mathbf{u}\cdot\mathbf{v})=(c\mathbf{u})\cdot\mathbf{v}$
4. $\mathbf{0}\cdot\mathbf{u}=0$
5. $\mathbf{u}\cdot\mathbf{u}=\|\mathbf{u}\|^2$

Theorem B

If θ is the smallest nonnegative angle between the nonzero vectors u and v, then

$$\mathbf{u}\cdot\mathbf{v}=\|\mathbf{u}\|\cdot\|\mathbf{v}\|\cos\theta$$

Theorem C Perpendicularity Criterion

Two vectors u and v are perpendicular if and only if their dot product $\mathbf{u}\cdot\mathbf{v}$ is 0.

Definition Orthogonal（正交的）

Vectors that are perpendicular are said to be orthogonal.

Remarks:

1. That $\vec{u}$ is orthogonal to $\vec{v}$ is denoted by $\vec{u}\perp\vec{v}$;

2. $\vec{0}\perp\vec{u},\forall\vec{u}$;

3. $\vec{u}\perp\vec{v}\Leftrightarrow\vec{u}\cdot\vec{v}=0\Leftrightarrow u_1v_1+u_2v_2+u_3v_3=0$;

4. $\vec{i}\cdot\vec{j}=\vec{j}\cdot\vec{k}=\vec{k}\cdot\vec{i}=0$,

5. $\lambda\vec{u}//\vec{u},\vec{0}//\vec{u},\forall\vec{u}$.

Direction Angles （方向角）and Cosines （方向余弦）：

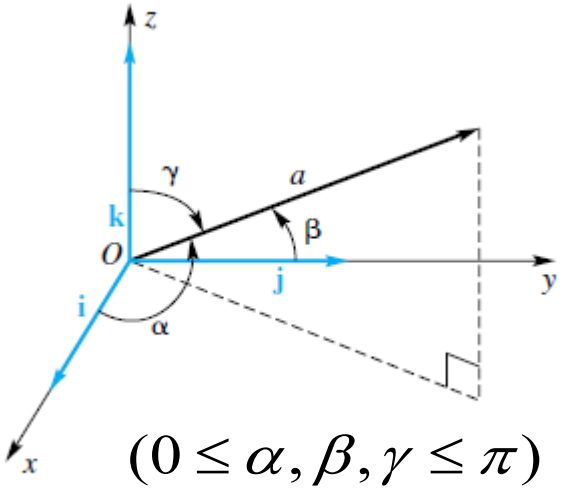
The smallest nonnegative angles between a nonzero three-dimensional vector **a** and the basis vectors **i**, **j**, and **k** are called the direction angles of **a**; they are denoted by α, β and γ, respectively, as shown in Figure 5. It is often more convenient to work with the direction cosines:

$\cos\alpha, \cos\beta$ and $\cos\gamma$. If $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$, then

$$\cos\alpha = \frac{\mathbf{a}\cdot\mathbf{i}}{\|\mathbf{a}\|\|\mathbf{i}\|} = \frac{a_1}{\|\mathbf{a}\|},$$

$$\cos\beta = \frac{\mathbf{a}\cdot\mathbf{j}}{\|\mathbf{a}\|\|\mathbf{j}\|} = \frac{a_2}{\|\mathbf{a}\|},$$

$$\cos\gamma = \frac{\mathbf{a}\cdot\mathbf{k}}{\|\mathbf{a}\|\|\mathbf{k}\|} = \frac{a_3}{\|\mathbf{a}\|},$$



$(0 \le \alpha, \beta, \gamma \le \pi)$

Figure 5

Notice that

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{a_1^2}{\|\mathbf{a}\|^2} + \frac{a_2^2}{\|\mathbf{a}\|^2} + \frac{a_3^2}{\|\mathbf{a}\|^2} = 1$$

Projections（投影） The **vector projection** of **u** on **v** or sometimes just the (vector) projection of **u** on **v**, is denoted

$\mathrm{pr}_{\mathbf{v}}\mathbf{u}(\text{or } \mathrm{proj}_{\mathbf{v}}\mathbf{u})$:

$$\mathrm{pr}_{v}\mathbf{u} = \left(\frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|^2}\right)\mathbf{v} = \left(\frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|}\right)\frac{\mathbf{v}}{\|\mathbf{v}\|}$$

The scalar projection of u on v is defined to be $\|\mathbf{u}\|\cos\theta$. Here θ is the angle between u and v. $(0 \le \theta \le \pi)$

Planes Let $\mathbf{n}=\langle A,B,C\rangle$ be a fixed nonzero vector and $P_1(x_1,y_1,z_1)$ be a fixed point. The set of points $P(x,y,z)$ satisfying $\overrightarrow{P_1P}\cdot\mathbf{n}=0$ is the plane through P_1 perpendicular to $\mathbf{n}$.

$\overrightarrow{P_1P}\cdot\mathbf{n}=0$ is equivalent to

$$A(x-x_1)+B(y-y_1)+C(z-z_1)=0$$

This equation (in which at least one of A, B, and C is different from zero) is called the standard form for the equation of a plane (平面方程的标准形式)

Linear equation of a plane:

$$Ax+By+Cz=D,\quad A^2+B^2+C^2\neq 0.$$

Example 10: Show that the distance L from the point (x_0, y_0, z_0) to the plane $Ax + By + Cz = D$ is given by the formula

$$L = \frac{|Ax_0 + By_0 + Cz_0 - D|}{\sqrt{A^2 + B^2 + C^2}}.$$

Distance between the parallel planes $Ax + By + Cz = D_1$
And $Ax + By + Cz = D_2$ is given by

$$L = \frac{|D_1 - D_2|}{\sqrt{A^2 + B^2 + C^2}}.$$

➢11.4 The Cross Product（叉积）

$$\boxed{\mathbf{u}\times\mathbf{v}=\begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ u_1 & u_2 & u_3\\ v_1 & v_2 & v_3\end{vmatrix}=\left(u_2v_3-u_3v_2\right)\mathbf{i}+\left(u_3v_1-u_1v_3\right)\mathbf{j}+\left(u_1v_2-u_2v_1\right)\mathbf{k}}$$

11.4 The Cross Product(叉积)

•The cross product $\mathbf{u}\times\mathbf{v}$ of $\mathbf{u}=\langle u_1,u_2,u_3\rangle$ and $\mathbf{v}=\langle v_1,v_2,v_3\rangle$ is defined by

$$\boxed{\mathbf{u}\times\mathbf{v}=\langle u_2v_3-u_3v_2,u_3v_1-u_1v_3,u_1v_2-u_2v_1\rangle}$$

Note that the cross product of two vectors is a vector.

Using determinants(行列式), we can write the definition of $\mathbf{u}\times\mathbf{v}$ as

$$\boxed{\mathbf{u}\times\mathbf{v}=\begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ u_1 & u_2 & u_3\\ v_1 & v_2 & v_3\end{vmatrix}=(u_2v_3-u_3v_2)\mathbf{i}+(u_3v_1-u_1v_3)\mathbf{j}+(u_1v_2-u_2v_1)\mathbf{k}}$$

Anti-commutative law(反交换律) $\mathbf{u}\times\mathbf{v}=-(\mathbf{v}\times\mathbf{u})$

Theorem A

1. $\mathbf{u}\cdot(\mathbf{u}\times\mathbf{v})=0=\mathbf{v}\cdot(\mathbf{u}\times\mathbf{v})$
2. $\mathbf{u},\mathbf{v},$ and $\mathbf{u}\times\mathbf{v}$ form a right-handed triple
3. $\|\mathbf{u}\times\mathbf{v}\|=\|\mathbf{u}\|\|\mathbf{v}\|\sin\theta$

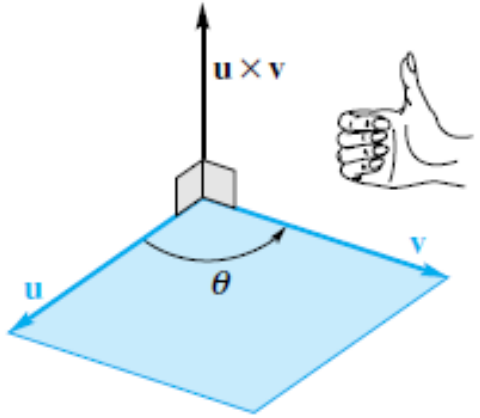


Figure 1

$\mathbf{u}\cdot\mathbf{v}$ and $\mathbf{u}\times\mathbf{v}$ are both intrinsic geometric quantities. You will get the same results for them no matter how you introduce the coordinates used to compute them.

Theorem C

If $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are vectors in three-space and k is a scalar, then

1. $\mathbf{u}\times\mathbf{v}=-(\mathbf{v}\times\mathbf{u})$
2. $\mathbf{u}\times(\mathbf{v}+\mathbf{w})=(\mathbf{u}\times\mathbf{v})+(\mathbf{u}\times\mathbf{w})$
3. $k(\mathbf{u}\times\mathbf{v})=(k\mathbf{u})\times\mathbf{v}=\mathbf{u}\times(k\mathbf{v})$
4. $\mathbf{u}\times\mathbf{0}=\mathbf{0}\times\mathbf{u}=\mathbf{0},\ \mathbf{u}\times\mathbf{u}=\mathbf{0}$
5. $(\mathbf{u}\times\mathbf{v})\cdot\mathbf{w}=\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w})$
6. $\mathbf{u}\times(\mathbf{v}\times\mathbf{w})=(\mathbf{u}\cdot\mathbf{w})\mathbf{v}-(\mathbf{u}\cdot\mathbf{v})\mathbf{w}$

Theorem B

Two vectors **u** and **v** in three-space are parallel if and only if $\mathbf{u}\times\mathbf{v}=\mathbf{0}$

Remarks:

$$\mathbf{u}=\langle u_1,u_2,u_3\rangle,\mathbf{v}=\langle v_1,v_2,v_3\rangle,$$

$$1.\mathbf{u}//\mathbf{v}\Leftrightarrow\exists\lambda,\mathbf{u}=\lambda\mathbf{v}\Leftrightarrow\frac{u_1}{v_1}=\frac{u_2}{v_2}=\frac{u_3}{v_3}$$

$$\Leftrightarrow\mathbf{u}\times\mathbf{v}=\mathbf{0}$$

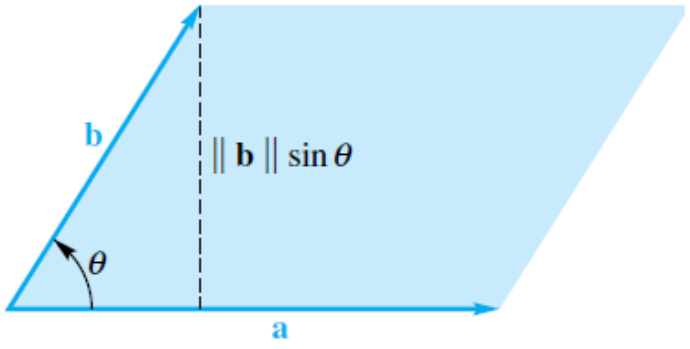
$$2.\mathbf{u}\perp\mathbf{v}\Leftrightarrow\mathbf{u}\cdot\mathbf{v}=0\Leftrightarrow u_1v_1+u_2v_2+u_3v_3=0$$

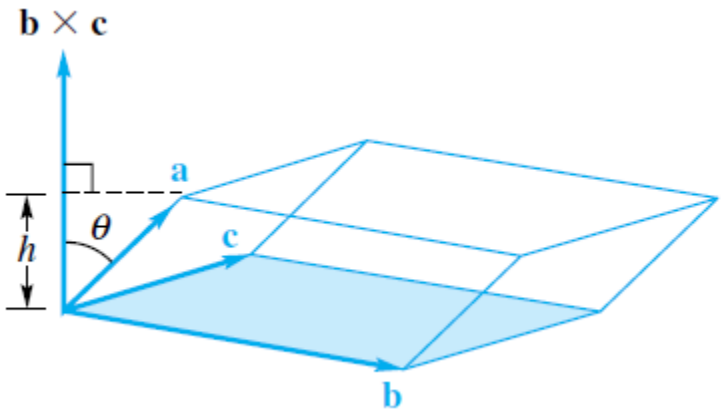
EXAMPLE 3 Show that the area of a parallelogram with **a** and **b** as adjacent sides is $\|\mathbf{a} \times \mathbf{b}\|$.

SOLUTION Recall that the area of a parallelogram is the product of the base times the height. Now look at Figure 3 and use the fact that $\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\|\|\mathbf{b}\| \sin \theta$. ■

EXAMPLE 4 Show that the volume of the parallelepiped determined by the vectors **a**, **b**, and **c** is

$$V = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})| = \left| \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \right|$$

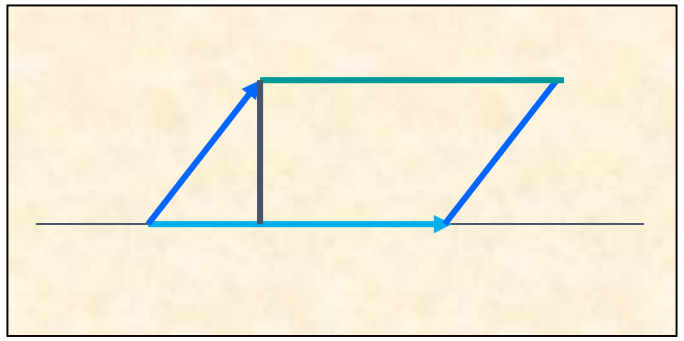




Ex.1 If the line l passes through points $A(1,1,1), B(1,-1,2)$

Find the distance from $C(2,1,1)$ to the line.

Sol. As shown in the figure,



$$\because \left|\overrightarrow{AB}\times\overrightarrow{AC}\right| = \left|\overrightarrow{AB}\right|\left|\overrightarrow{AC}\right|\sin\theta$$

$$= \left|\overrightarrow{AB}\right| d$$

$$\therefore d = \frac{\left|\overrightarrow{AB}\times\overrightarrow{AC}\right|}{\left|\overrightarrow{AB}\right|},$$

$$\because \quad \overrightarrow{AB} = (0,-2,1), \quad \overrightarrow{AC} = (1,0,0), \quad \overrightarrow{AB}\times\overrightarrow{AC} = (0,1,2),$$

$$\Rightarrow \left|\overrightarrow{AB}\right| = \left|\overrightarrow{AB}\times\overrightarrow{AC}\right| = \sqrt{5},$$

$$\therefore \quad d = 1.$$

- 11.5 Vector-Valued Functions

11.5 Vector-Valued Functions and Curvilinear Motion

- **Vector-Valued Functions**

A vector-valued function **F** of a real variable *t* associates with each real number *t* a unique vector **F**(*t*). Thus

$$\mathbf{F}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k} = \langle f(t), g(t), h(t) \rangle$$

where *f*, *g* and *h* are real-valued functions.

Calculus for Vector Functions

Theorem A

Let $\mathbf{F}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$. Then $\mathbf{F}$ has a limit at c if and only if $f, g,$ and h have limits at c. In this case,

$$\lim_{t \to c} \mathbf{F}(t) = \left[\lim_{t \to c} f(t)\right]\mathbf{i} + \left[\lim_{t \to c} g(t)\right]\mathbf{j} + \left[\lim_{t \to c} h(t)\right]\mathbf{k}$$

- **Derivative of a Vector-Valued Function**

If $\mathbf{F}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, then

$$\mathbf{F}'(t) = f'(t)\mathbf{i} + g'(t)\mathbf{j} + h'(t)\mathbf{k} = \langle f'(t), g'(t), h'(t) \rangle$$

- **Integral of a Vector-Valued Function**

If $\mathbf{F}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, it is natural to define integration in terms of components.

$$\int \mathbf{F}(t)dt = \left[\int f(t)dt\right]\mathbf{i} + \left[\int g(t)dt\right]\mathbf{j} + \left[\int h(t)dt\right]\mathbf{k}$$

$$\int_a^b \mathbf{F}(t)dt = \left[\int_a^b f(t)dt\right]\mathbf{i} + \left[\int_a^b g(t)dt\right]\mathbf{j} + \left[\int_a^b h(t)dt\right]\mathbf{k}$$

- **Curvilinear Motion(曲线运动)**

Let t measure time, and suppose that the coordinates of a moving point P in space are given by the parametric equations $x = f(t), y = g(t), z = h(t)$. Then the vector

$$\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$$

assumed to emanate from the origin, is called the ***position vector*** of the point. As t varies, the head of $\mathbf{r}$(t) traces the path of the moving point P (Figure 4). This is a curve, and we call the corresponding motion ***curvilinear motion***.

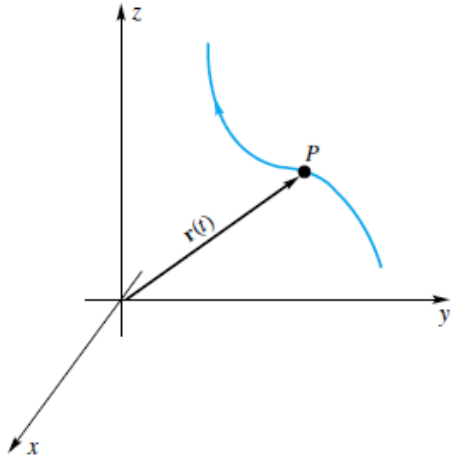


Figure 4

The ***velocity*** $\mathbf{v}(t)$ and the ***acceleration*** $\mathbf{a}(t)$ of the moving point P is defined by

$$\mathbf{v}(t) = \mathbf{r}'(t) = f'(t)\mathbf{i} + g'(t)\mathbf{j} + h'(t)\mathbf{k}$$

$$\mathbf{a}(t) = \mathbf{r}''(t) = f''(t)\mathbf{i} + g''(t)\mathbf{j} + h''(t)\mathbf{k}$$

The accumulated ***arc length*** from time $t = a$ to an arbitrary time t is thus

$$\boxed{s = \int_a^t \sqrt{[f'(u)]^2 + [g'(u)]^2 + [h'(u)]^2}\, du = \int_a^t \|\mathbf{r}'(u)\|\, du}$$

Its derivative (i.e. ***speed***) is

$$\frac{ds}{dt} = \sqrt{[f'(u)]^2 + [g'(u)]^2 + [h'(u)]^2} = \|\mathbf{r}'(t)\|$$

- 11.6 Line and Tangent Line in Three Space

- **Equations for Lines**

$$\boxed{x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct}$$

- These are called ***parametric equations*** of the line。
- (x_0, y_0, z_0) is a point on the line
- $\mathbf{v} = \langle a, b, c \rangle$ is the directional vector of the line . *a, b, c* are called ***directional numbers*** for the line.

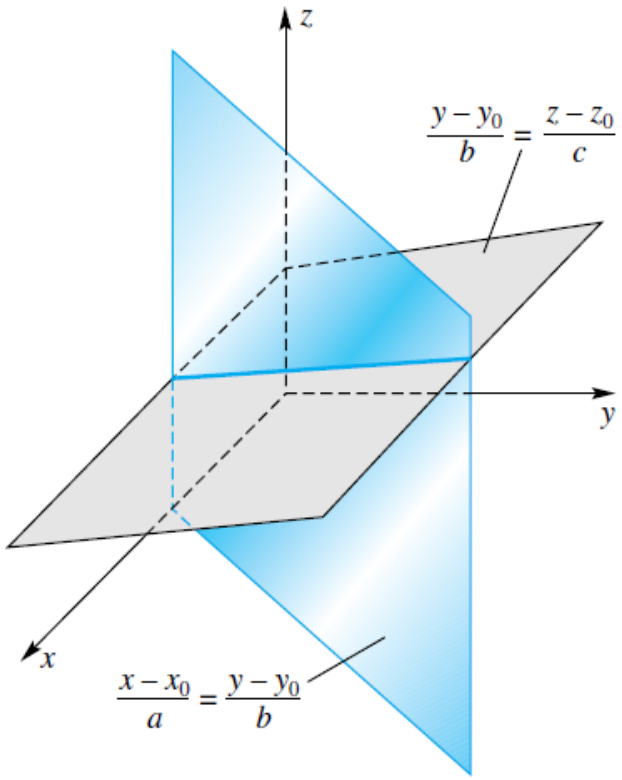
From the parametric equations we can get the ***symmetric equations*** *(对称方程)* for line through (x_0, y_0, z_0) with direction numbers a, b, c

$$\frac{x-x_0}{a}=\frac{y-y_0}{b}=\frac{z-z_0}{c}$$

This line can be regarded as the intersection line of the two planes:

$$\frac{x-x_0}{a}=\frac{y-y_0}{b}$$

and $$\frac{y-y_0}{b}=\frac{z-z_0}{c}$$



- **Tangent Line to a Curve**

Let $\mathbf{r} = \mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k} = \langle f(t), g(t), h(t) \rangle$

be the position vector determining a curve in three-space (Figure 5).

The tangent line to the curve has direction vector

$$\mathbf{r}'(t) = f'(t)\mathbf{i} + g'(t)\mathbf{j} + h'(t)\mathbf{k} = \langle f'(t), g'(t), h'(t) \rangle$$

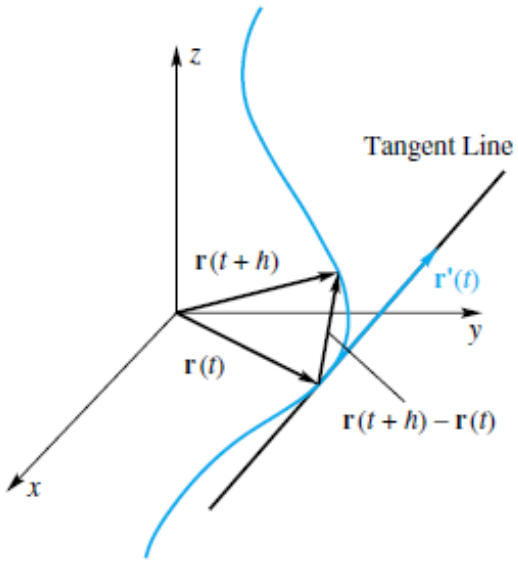


Figure 5

11.7 Curvature(曲率)

A curve in 3-space can be represented by parametric equation

$$\begin{cases} x = f(t) \\ y = g(t), a < t < b \\ z = h(t) \end{cases}$$

or by corresponding position vector（位置向量）

$$\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k} = \langle f(t), g(t), h(t) \rangle,\ a < t < b$$

We assume that $\mathbf{r}'(t)$ is continous and that $\mathbf{r}'(t) \neq \mathbf{0}$.

We also assume that the arc length $s(t)$ increases as t increases.

The tangent vector（切向量）at $(f(t), g(t), h(t))$ is $\mathbf{r}'(t)$;

The tangent vector is the direction vector of the tangent line;

The unit tangent vector thus is $\mathbf{T}(t) = \dfrac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}$.

Definition

The curvature κ is the magnitude of the rate of change of the unit tangent vector with respect to arc length s:

$$\kappa = \left\| \frac{d\mathbf{T}}{ds} \right\|$$

$$\frac{d\mathbf{T}}{ds} = \frac{d\mathbf{T}}{dt}\frac{dt}{ds} = \frac{\frac{d\mathbf{T}}{dt}}{\frac{ds}{dt}} = \frac{\mathbf{T}'(t)}{\|\mathbf{r}'(\mathbf{t})\|}, \text{ where speed} = \|\mathbf{v}(t)\| = \frac{ds}{dt} = \|\mathbf{r}'(t)\|$$

$$\boxed{\text{Curvature } \kappa = \left\| \frac{d\mathbf{T}}{ds} \right\| = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(\mathbf{t})\|}.}$$

Radius and Center of Curvature for a Plane Curve

Circle of Curvature(曲率圆):

The circle that is tangent to the curve and has the same curvature there

Radius of curvature(曲率半径):

$$R = 1/\kappa$$

Center of Curvature(曲率中心):

the center of the circle of curvature

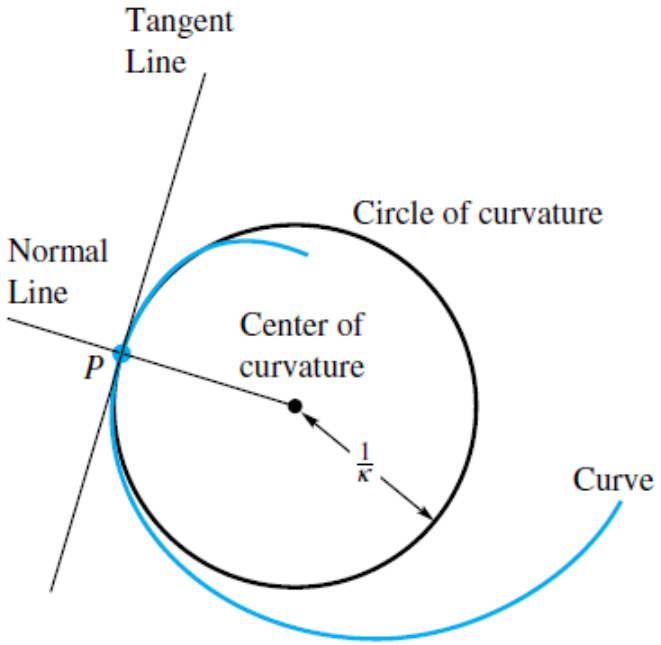


Figure 6

- Other Formulas for Curvature of a Plane Curve

Theorem

Consider a curve with vector equation $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j}$, that is, with parametric equation $x = f(t)$ and $y = g(t)$. Then

$$\kappa = \frac{|x'y'' - y'x''|}{\left[(x')^2 + (y')^2\right]^{3/2}}.$$

In particular, if the curve is the graph of $y = g(x)$, then

$$\kappa = \frac{|y''|}{\left[1 + (y')^2\right]^{3/2}}$$

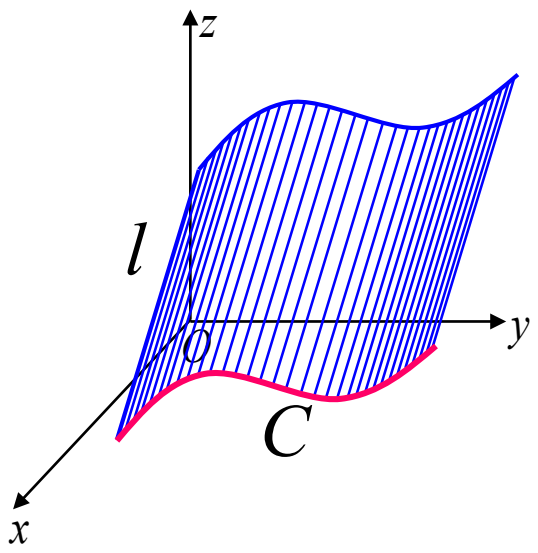
- 11.8 Surfaces in Three-Space

- **Cylinders**

A simple description of cylinders:

A *cylinder* is a surface generated by parallel moving a given line l along a plane curve C.

The moving line l is called the generator of the cylinder.

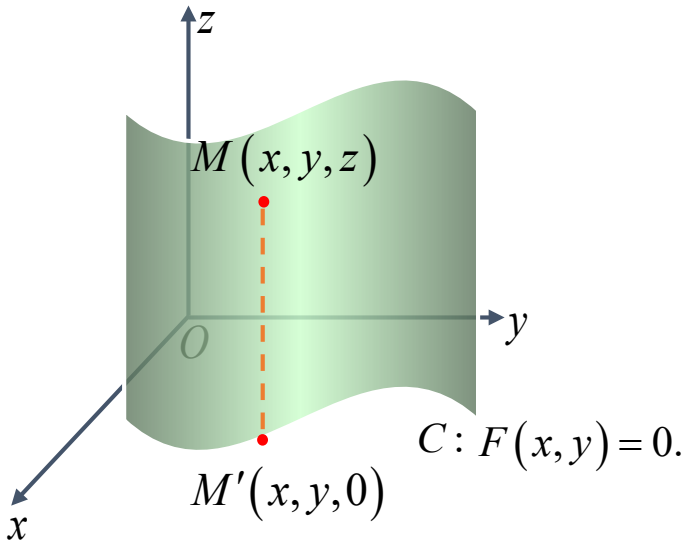


Equations for Cylinders

If a cylinder Σ satisfies: the given line l is parallel to z-axis; the given curve C is the plane curve $F(x,y)=0$ in the xy-plane.

Therefore, the equation of the cylinder Σ is also

$$\boxed{F(x,y)=0.}$$



In general, for $F(x, y, z) = 0$.

If one of the variables x, y or z is missing in the equation, then the graph of the equation is a **cylinder whose generator is parallel to the axis of that missing variable**.

Thus,

$F(x, y) = 0$ represents a cylinder whose generator is parallel to z-axis;

$G(y, z) = 0$ represents a cylinder whose generator is parallel to x-axis;

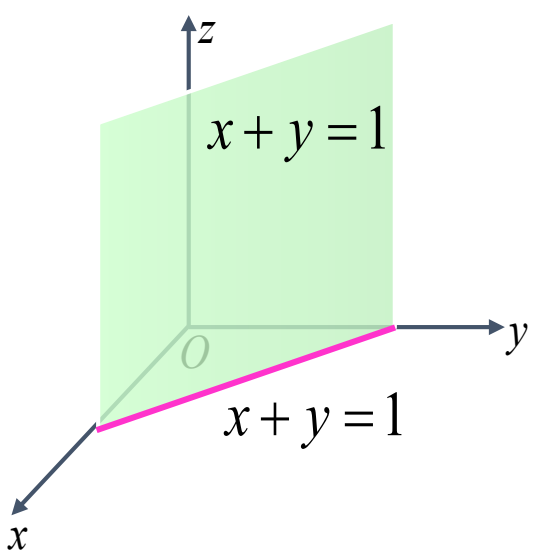
$H(z, x) = 0$ represents a cylinder whose generator is parallel to y-axis.

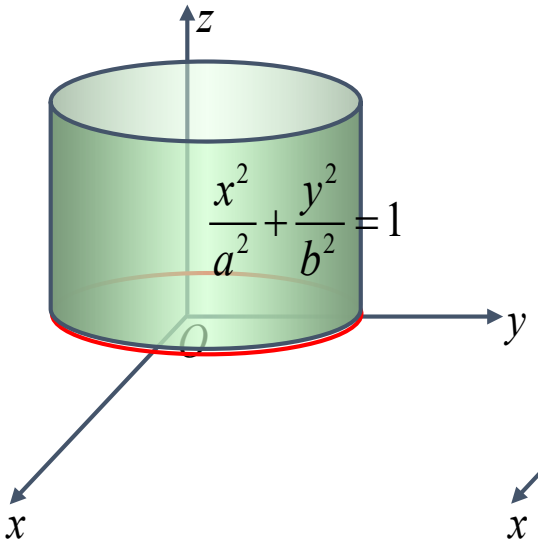
Examples

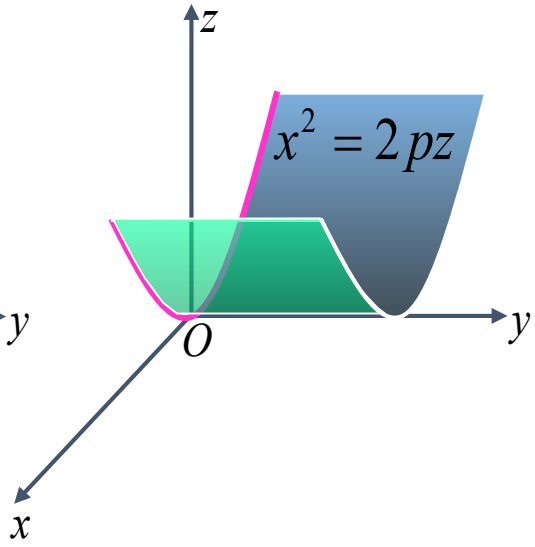
Ex 1 The graph of $x+y=1$ is a plane parallel to the z-axis. This plane can be regarded as a special case of a cylinder.

Ex 2 The graph of $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is an **elliptic cylinder** whose generator is parallel to the z-axis.

Ex 3 The graph of $x^2=2pz$ is a **parabolic cylinder** whose generator is parallel to y-axis.







3. Quadric Surfaces

A **quadric surface** is the graph in three-space of a second-degree equation in three variables *x*, *y* and *z*.

The most general second-degree equation has the form

$$Ax^2 + By^2 + Cz^2 + Dxy + Exz + Fyz + Gx + Hy + Iz + J = 0$$

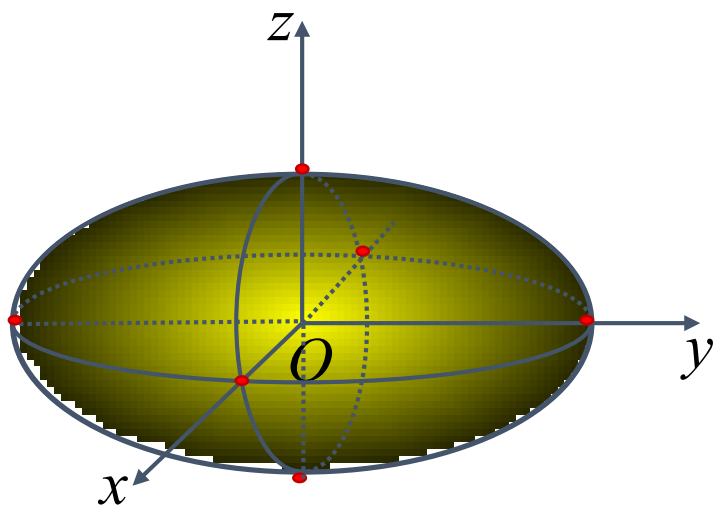
By rotation and translation of coordinate axes, such equation can be reduced to one of the two standard forms

$$Ax^2 + By^2 + Cz^2 + J = 0 \quad \text{or} \quad Ax^2 + By^2 + Iz = 0$$

We will show you **six** general types of quadric surfaces.

1. Ellipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1, \quad (a>0, b>0, c>0).$

Sketch the graph of $\frac{x^2}{4}+\frac{y^2}{9}+\frac{z^2}{1}=1.$



1. Find the x, y, z-intercepts:

   set $y=z=0$, then $x=\pm 2$;

   set $x=z=0$, then $y=\pm 3$;

   set $x=y=0$, then $z=\pm 1$.

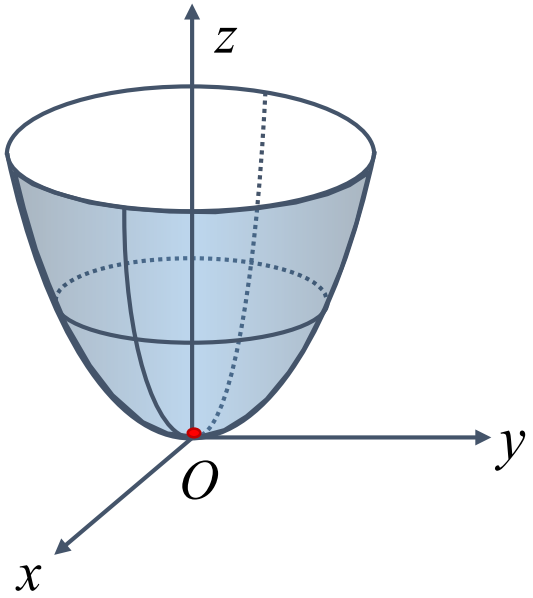
2. Find the traces in the coordinate planes:

   set $x=0$, then $\frac{y^2}{9}+\frac{z^2}{1}=1$, an ellipse in the yz-plane;

   set $y=0$, then $\frac{x^2}{4}+\frac{z^2}{1}=1$, an ellipse in the yz-plane;

   set $z=0$, then $\frac{x^2}{4}+\frac{y^2}{9}=1$, an ellipse in the yz-plane.

2. Elliptic Paraboloid $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=z.$



Sketch the graph of $\dfrac{x^2}{4}+\dfrac{y^2}{1}=z.$

set $z=0$, then $x=y=0$, a point in the xy-plane;

set $z=1$, then $\dfrac{x^2}{4}+\dfrac{y^2}{1}=1,$ an ellipse in the plane $z=1$;

set $x=0$, then $y^2=z$, a parabola in the yz-plane.

set $y=0$, then $\dfrac{x^2}{4}=z$, a parabola in the xz-plane.

3. Hyperbolic Paraboloid $\dfrac{x^2}{a^2}-\dfrac{y^2}{b^2}=z.$

Sketch the graph of $\dfrac{x^2}{4}-\dfrac{y^2}{1}=z.$

set $z=0$, then $y=\pm\dfrac{x}{2}$, two intersection lines in the xy-plane;

set $z=1$, then $\dfrac{x^2}{4}-\dfrac{y^2}{1}=1$, a hyperbola in the plane $z=1$;

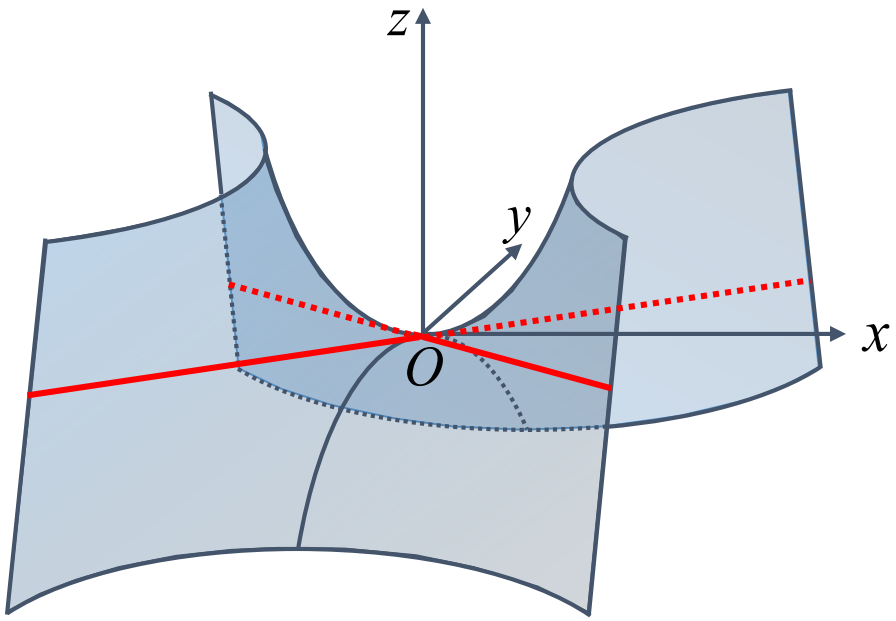
set $z=-1$, then $\dfrac{y^2}{1}-\dfrac{x^2}{4}=1$, a hyperbola in the plane $z=-1$;

set $x=0$, then $-y^2=z$,

a parabola in the yz-plane ;

set $y=0$, then $\dfrac{x^2}{4}=z$,

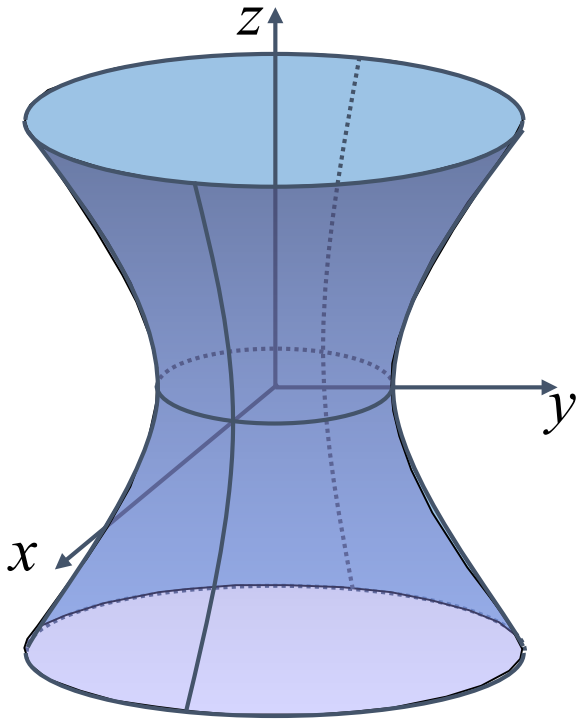
a parabola in the xz-plane ;



4. Hyperboloid of One Sheet $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}-\dfrac{z^2}{c^2}=1.$

Exercise: Sketch the graph of

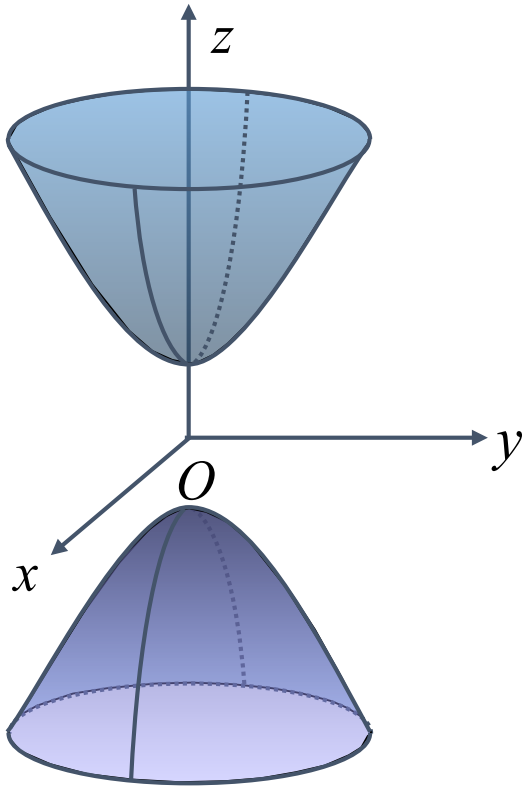
$$x^2+y^2-\frac{z^2}{4}=1.$$



5. Hyperboloid of Two Sheets $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}-\dfrac{z^2}{c^2}=-1.$

Exercise: Sketch the graph of

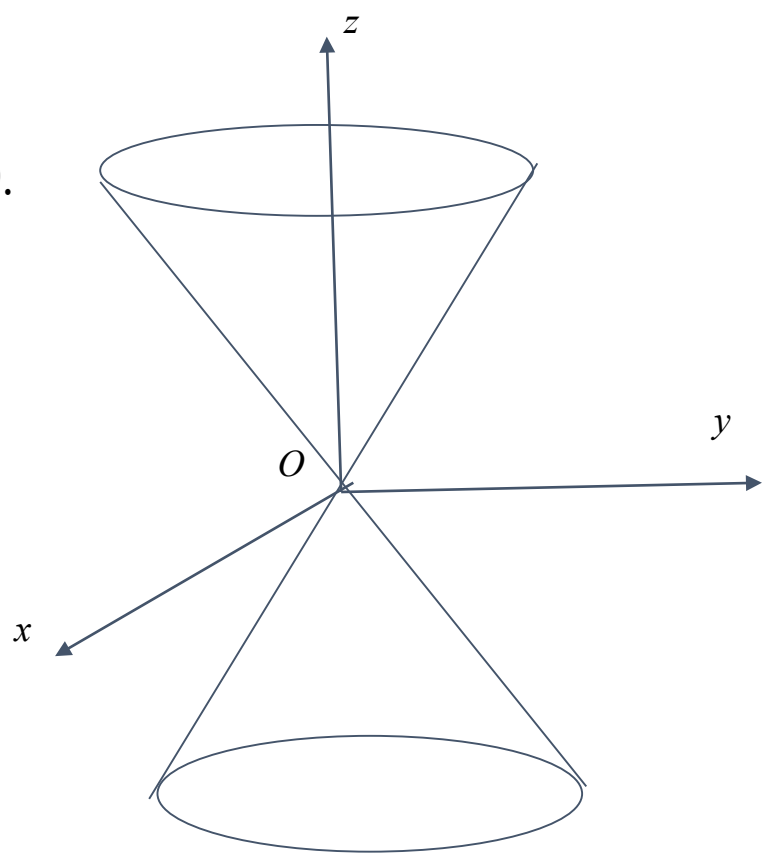
$$x^2+\frac{y^2}{2}-z^2=-1.$$



6. Elliptic Cone $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}-\dfrac{z^2}{c^2}=0$

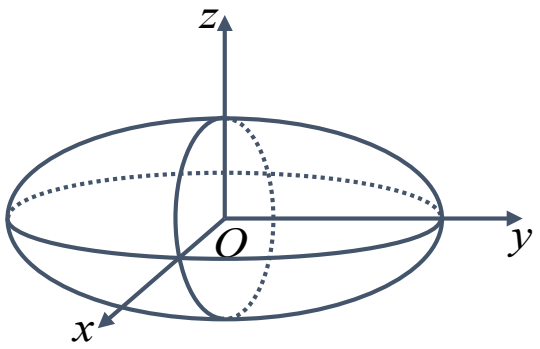
Exercise: Sketch the graph of

$$x^2+\frac{y^2}{2}-z^2=0.$$

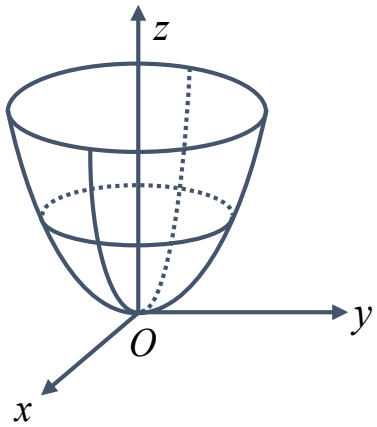


Summary

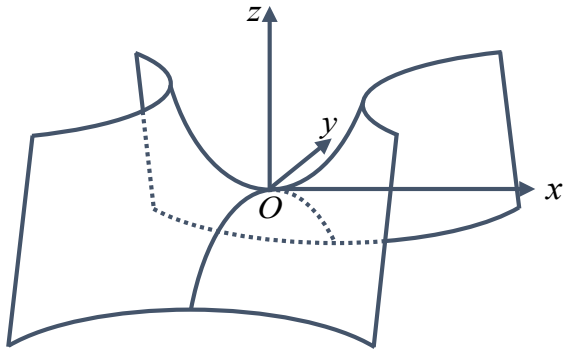
1. **Ellipsoid** $$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$$



2. **Elliptic Paraboloid** $$\frac{x^2}{a^2}+\frac{y^2}{b^2}=z$$

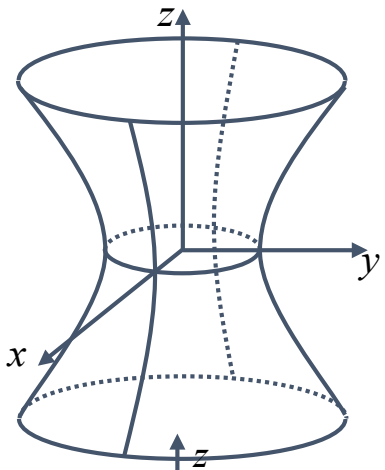


3. **Hyperbolic Paraboloid** $$\frac{x^2}{a^2}-\frac{y^2}{b^2}=z$$



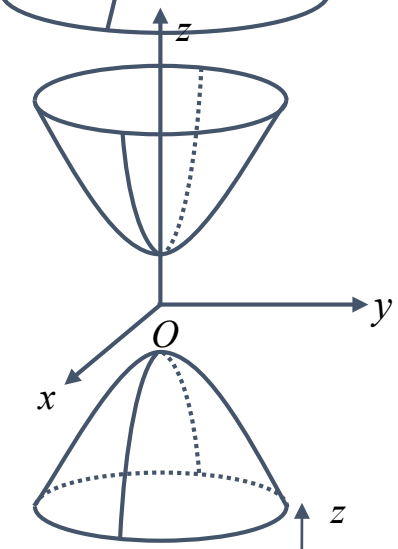
4. Hyperboloid of One Sheet

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=1$$



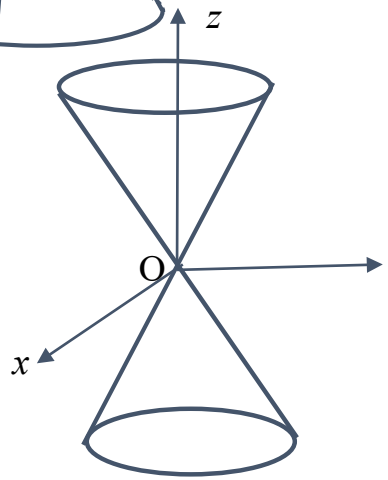
5. Hyperboloid of Two Sheets

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=-1 \Leftrightarrow -\frac{x^2}{a^2}-\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$$



6. Elliptic Cone

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=0$$

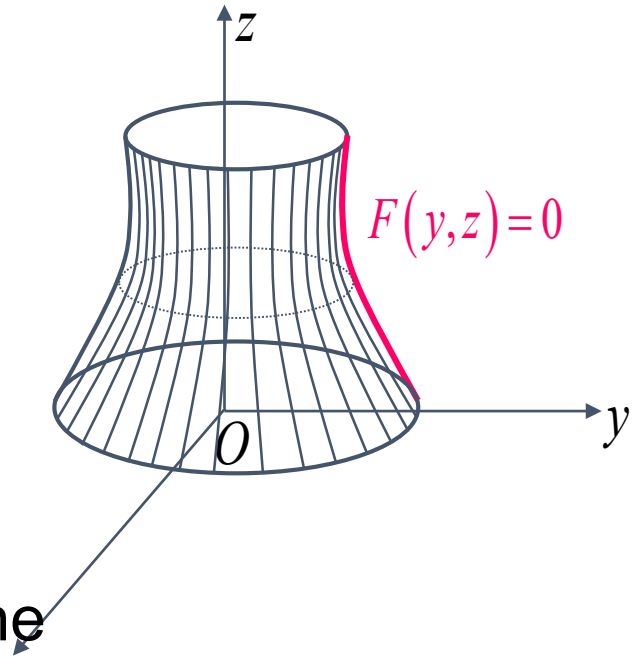


3.Surface of revolution（旋转曲面）

Now suppose C be a curve on yOz plane, whose 2-dimensional equation is $F(y,z)=0.$

The axis of revolution is z-axis.

$$\boxed{F\left(\pm\sqrt{x^2+y^2},z\right)=0,}$$

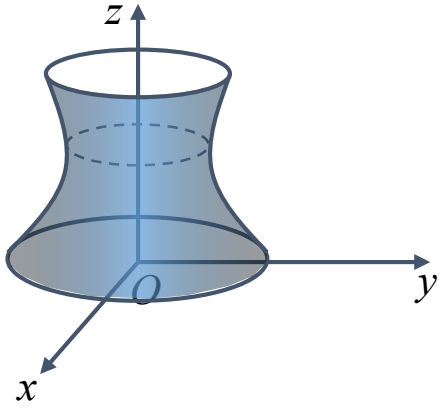


In generator curve F(**y**,z)=0, you replace the variable that is **NOT** the axis of revolution y by plus-minus square root of the square of x which is the variable not showing up in F(y,z)=0 plus the square of y which is the variable that is replaced in F(y,z)=0.

Similarly, we may conclude:

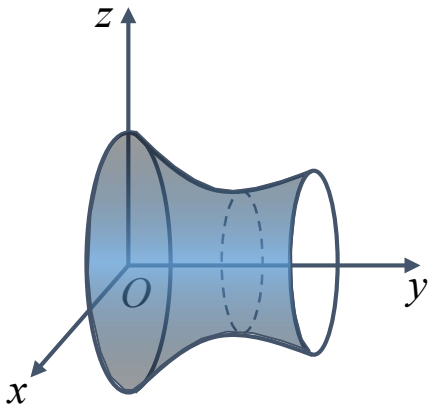
(1) The equation of the surface generated by revolving the curve $F(y,z)=0$ in the *yz*-plane about the *z*-axis is

$$\boxed{F\left(\pm\sqrt{x^2+y^2},z\right)=0,}$$



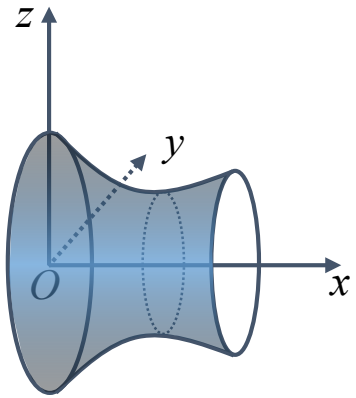
(2) The equation of the surface generated by revolving the curve $F(y,z)=0$ in the *yz*-plane about the *y*-axis is

$$\boxed{F\left(y,\pm\sqrt{x^2+z^2}\right)=0,}$$



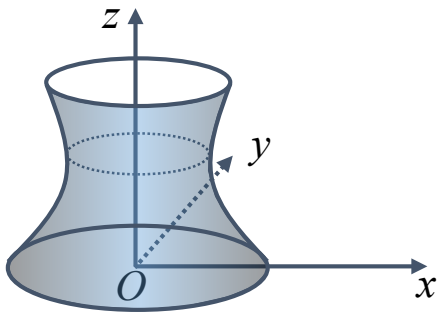
(3) The equation of the surface generated by revolving the curve $G(x,z)=0$ in the *xz*-plane about the *x*-axis is

$$\boxed{G\left(x,\pm\sqrt{y^2+z^2}\right)=0,}$$



(4) The equation of the surface generated by revolving the curve $G(x,z)=0$ in the *xz*-plane about the *z*-axis is

$$\boxed{G\left(\pm\sqrt{x^2+y^2},z\right)=0,}$$



Question: How about revolving the curve $H(x, y) = 0$ in the *xy*-plane about the *x*-axis?

$$H(x, \pm\sqrt{y^2 + z^2}) = 0$$

Question: How about revolving the curve $H(x, y) = 0$ in the *xy*-plane about the *y*-axis?

$$H(\pm\sqrt{x^2 + z^2}, y) = 0$$